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TOPICAL REVIEW

A Review of Large Language Models for Energy Systems: Applications, Challenges, and Future Prospects

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ABSTRACT Rapid developments in large language models (LLMs) have created new opportunities for their use in the energy sector, from forecasting renewable energy to power system operation and energy market analysis. These models help improve decision-making, anomaly detection, and optimization procedures in intricate energy systems by using vast amounts of structured and unstructured data. This study provides a comprehensive review of the LLM origins, evaluation, and fine-tuning techniques as well as their integration into energy systems, including their application in fault detection and diagnosis, energy forecasting, document automation, energy management, defect detection, and power system operation. Their performance in terms of explainability, generalization ability, and scalability for energy-related applications is critically examined in this paper. The report also emphasizes significant challenges to the adoption of LLMs, such as the need for computing power, the lack of data, and ethical issues like bias and false information. Power-efficient models, hybrid artificial intelligence (AI) platforms, and domain-specific fine-tuning are some of the solutions discussed. Future areas of interest include multi-modality to obtain maximal forecasting and operational intelligence, real-time adaptability, and explainable. This paper summarizes current developments and provides information on LLM-driven innovation in energy systems while maintaining transparency and dependability. Compared with prior LLM–energy surveys that either remain general-purpose or focus on a single subdomain, this review fills three concrete gaps: (i) a cross-domain synthesis of *energy-specific* LLM applications spanning power systems, buildings, and forecasting; (ii) a methods-oriented consolidation of evaluation and parameter-efficient fine-tuning practices tailored to energy tasks; and (iii) a deployment-centric analysis of real-time and edge constraints (energy, latency, hardware) with a practical reporting checklist for operational adoption.

INDEX TERMS Large language models, energy systems, fault detection and diagnosis, explainable artificial intelligence, energy forecasting.

I. INTRODUCTION

The rising penetration of renewable energy resources, decentralized generation, and the mounting need for smart grid technologies are fundamentally changing the nature

of energy systems. The variable and unstable nature of renewable generation, combined with the complexity of electricity markets, has given rise to several challenges related to system reliability, demand-supply balance, and operational efficiency [1]. The conventional methods, including statistical prediction, rule-based optimization, and traditional physics-based modeling, have been used to resolve these

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issues [2]. However, when the energy system becomes more complex and dynamic, together with its interaction with other forms of energy systems, the scaling efficiency of traditional techniques is compromised by real-time changes [3].

AI has emerged as a transformative solution in energy management, fault detection, and forecasting, offering data-driven insights that enhance decision-making and system resilience [4], [5]. Machine learning and deep learning-based models applied to renewable energy forecasting, optimal market operation, and predictive maintenance have promising applications [6]. Nevertheless, these developments make conventional AI approaches require enormous labeled datasets, domain-specific retraining, and significant amounts of computational power, which limit their generalization and scalability potential in real-life energy-related applications [7].

In recent years, the emergence of LLMs has revolutionized AI applications across various domains. LLMs such as GPT-4, BERT, and T5 are deep learning-based models that are trained on massive amounts of text data, enabling them to understand, generate, and reason over human language with great accuracy [8]. Unlike traditional AI models, which depend mostly on structured numerical datasets, LLMs have the capability to process both structured and unstructured data and are therefore extremely versatile in handling intricate real-world applications, such as energy systems [9]. LLMs' main strength lies in their ability to condense knowledge from large corpora of text data, including scientific articles, technical reports, operational data, and policy documents, thereby providing a powerful tool for knowledge retrieval and contextual reasoning.

These models can be fine-tuned for specific tasks such as fault diagnosis, demand prediction, and market prediction with significantly less need for large-scale domain-specific retraining [10]. Advances in retrieval-augmented generation (RAG), transfer learning, and multi-modal AI have also expanded the application of LLMs in processing multi-source data such as time-series sensor data, maintenance records, and text reports [11]. In the context of energy systems, LLMs offer unique advantages in automating decision-making, optimizing grid operations, and enhancing system reliability.

Their ability to process natural language requests makes them beneficial to human-AI interaction in control rooms, with operators being capable of retrieving major insights as well as anomaly detection in real time [12]. LLMs can also be integrated with machine learning pipelines to enhance forecasting accuracy and provide intelligent recommendations for grid optimization [13]. As energy systems become increasingly complex and data-driven, LLMs will play a key role in advancing operational intelligence and sustainability.

This paper provides a general overview of the applications, challenges, and future directions of LLMs in energy systems, focusing on their applications in power system operation, building energy management, and time-series forecasting of energy-related variables. The primary objectives of this work are:

- To provide a review of LLM fundamentals, evaluation metrics, fine-tuning strategies, and energy-specific models based on LLM.
- To investigate LLM applications across major energy domains, including:
 - Power system applications such as fault diagnosis, grid operation, market participation, and document automation.
 - Building energy systems including HVAC fault detection, smart control, and modeling.
 - Energy forecasting tasks such as load prediction, electricity price forecasting, and renewable energy (PV and wind) forecasting.
- To discuss the current challenges—such as high computational cost, data scarcity, and trustworthiness—in deploying LLMs in energy domains, and to explore potential solutions to address them.

Table 1 offers a thorough summary of the present review article, together with other state-of-the-art studies in the field to make comparison easier. The table provides a clear overview of the distinctive contributions and methodologies of each publication by highlighting important parallels and distinctions. Despite several recent surveys on LLMs and energy, clear gaps remain. Most works either provide broad AI overviews with limited energy-specific modeling detail or concentrate on a single task/domain (e.g., forecasting *or* power-system operation), offering little guidance on (a) parameter-efficient fine-tuning and evaluation protocols customized for energy data; (b) cross-domain taxonomies and timelines of *energy-specific* LLM tools; and (c) the practical costs of deployment under real-time and edge constraints (energy use, latency budgets, hardware limits) [1], [10], [14], [15], [16], [17], [18]. This review addresses those gaps by (i) synthesizing applications across power systems, buildings, and forecasting with emphasis on *energy-specific* LLM designs; (ii) systematizing evaluation and fine-tuning choices shown to be effective in energy settings; and (iii) introducing a deployment-oriented analysis and reporting checklist that makes capability–cost trade-offs explicit for meters, DER controllers, and microgrids.

II. LITERATURE REVIEW METHODOLOGY AND KEYWORD ANALYSIS

The present study employs the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) methodology to review systematically how LLMs have been utilized in energy systems. It systematically searched prominent research databases, using keyword analysis and research topics to screen the relevant publications.

A. KEYWORD CO-OCCURRENCE ANALYSIS

To investigate thematic links between LLM uses, a bibliometric analysis was performed. Keyword frequencies in article titles and abstracts are visualized in Figure 1. It depicts the extensive usage of LLMs in power system operation, fault

TABLE 1. Comparison of this study with other state-of-the-art review papers.

Feature	This study	[1]	[14]	[15]	[16]	[10]	[17]	[18]
LLM Eval & PEFT (metrics or concrete PEFT)	Eval+PEFT	Persp.; eval	no new	Not LLM	SLR: tasks+met.	TS-FM eval	Review; LoRA	Comm.; no Eval survey met.
Energy-specific LLMs/Agents (introduces named energy LLM/agent/fw)	Domain LLMs cat.	No model	new	None	None	TS-FM (not LLM)	Reviews; no model	RAG/tools, none
Power System (grid methods/datasets/focused section)	Dedicated	Building-centric	BTM/DER focus	Limited energy	LF+overload	Energy incl. wind	OPF/tools note	Not domain
Building Energy Modeling (BEM) (EPlus/BMS workflows)	BEM surveyed	BEM/EPlus opp.	Not BEM	—	—	EPlus-LLM rev.	Not BEM	—
Load Forecasting (explicit LF models/exp. or review)	LF compared	Cites only	BTM LF+met.	Generic TS	LF eval	Bldg LF+LLMs	Not LF rev.	—
PV & Wind Forecasting (explicit PV/wind models/review)	PV+Wind	—	PV (BTM); wind lim.	—	—	Wind (STELLM)	—	—
Electricity Price Forecasting (explicit price models/exp./review)	Price covered	—	—	—	—	Not explicit	Mkts/OPF; not price	—
FDD in Energy Systems (HVAC/power-asset faults with LLM role)	HVAC+grid FDD	Control use-cases	BTM anom. (non-HVAC)	Anom. (generic)	Overload (as-set)	HVAC FDD	FDD	Not FDD

Abbrev.: PEFT=parameter-efficient fine-tuning; TS-FM=time-series foundation model; BTM=behind-the-meter; BEM=building energy modeling; LF=load forecasting; OPF=optimal power flow; rev.=review; met.=metrics; fw=framework.

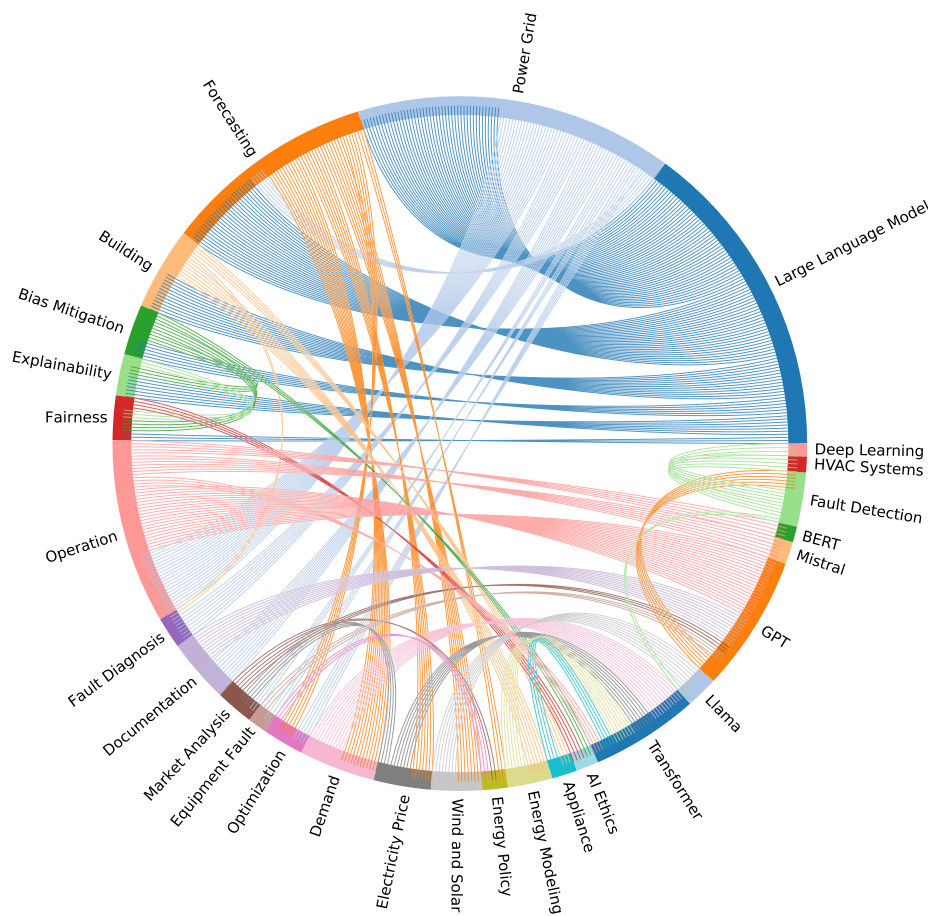


FIGURE 1. Visualization of keyword search co-occurrences across referenced papers.

diagnosis, forecasting, and energy policy. The analysis is aligned with studies on AI-informed energy optimization, maintaining the predictive modeling and decision-making capability of LLMs.

B. CATEGORIZATION OF LLM APPLICATIONS IN ENERGY SYSTEMS

To put the contributions of LLMs into perspective, Figure 2 categorizes the studies discussed based on the application

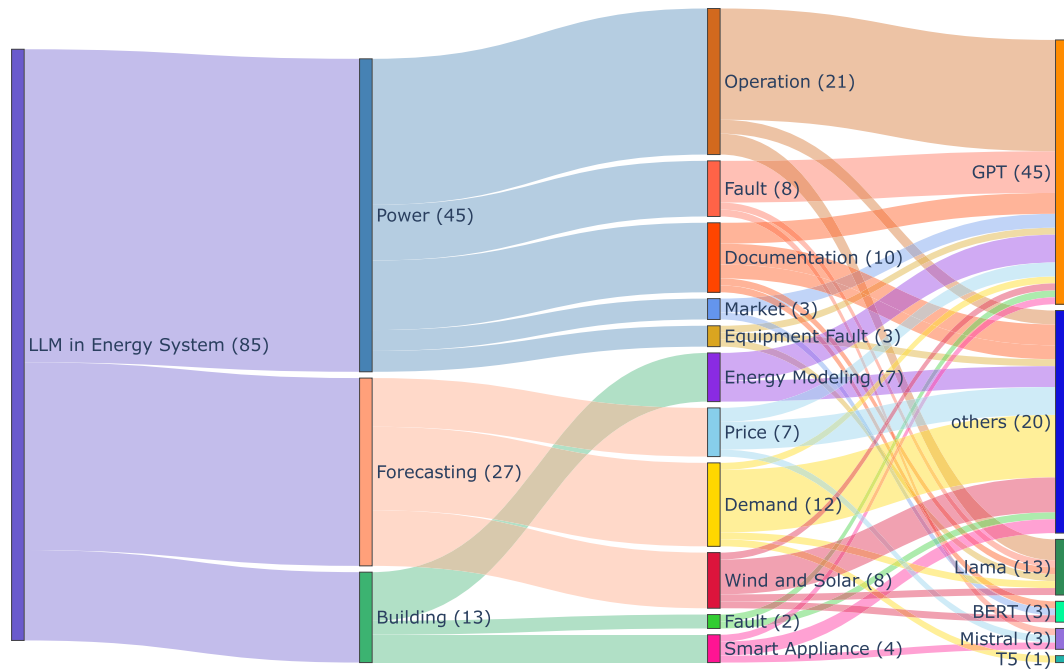


FIGURE 2. Publications categorized by LLM application in energy systems and LLM models used.

domain and respective LLM models used. Three broad categories are identified through examination: power systems (fault detection, market forecasting), building energy systems (HVAC control, energy modeling), and forecasting (load, price, and renewable energy forecasting).

These LLM models employed are GPT, BERT, Llama, T5, Mistral, and similar transformer-based models. The vast majority of models in the literature utilized for code generation, optimization, interaction, and other purposes in the energy-specific domain are GPT-based.

The findings emphasize the increasing contribution of LLM to energy research, providing insights into future trends and potential innovations in AI-driven energy systems.

III. LARGE LANGUAGE MODELS

A. LLM DEFINITION

An LLM is a subset of AI that enables a program to interact with humans and produce text using vast amounts of data and advanced neural networks [19]. One of the most crucial elements of recently developed models is the enormous number (billions) of parameters, which assist in parameter modeling and capture complex speech structures, contextual information, and semantics. LLMs have sequential prediction of the next word in a sentence as their core function. This is made possible by large-scale datasets that significantly differ from one another in how these LLMs are trained. This pre-training stage incorporates self-supervised approaches in the model using various resources such as books, the internet, and other text resources. Its major benefit is that it is label-free, meaning it is not reliant on any human-associated labels [20]. The scalability and generalizability of LLMs allow them to

be deployed for various tasks in different domains with very little task-specific training required.

Recent works further sharpen the definition of LLMs: for instance, representation- and contrastive-learning approaches explicitly target multimodal hallucination reduction during training (e.g., Hallucination-Augmented Contrastive Learning) [21], while decoding-time and data-centric methods suppress factual errors in noisy or domain-specific contexts [22], [23].

These methods typically report progress on contemporary hallucination and faithfulness benchmarks (e.g., MMHal-Bench, explainable QA suites), emphasizing that modern LLM definitions increasingly bundle raw generative skill with robustness to adversarial or noisy evidence [21], [24]. For free-form generation, semantic-entropy estimators can flag confabulations without task-specific tuning, improving answer triage across QA datasets [25]; these detection signals complement training-time contrastive objectives and decoding-time strategies that target hallucinations and grounding [21], [22], [23].

B. EVOLUTION OF LLMs

1) EARLY MODELS

The development of LLMs is based on the pioneering work that was done with simpler neural network models, which allowed the growth of complex systems as we know them today. Among the first attempts to model sequential data were the so-called Recurrent Neural Networks. First introduced in [26], RNNs utilize recurrent connections to process sequences by maintaining a hidden state, capturing

information from previous states. However, RNNs had limitations in modeling long-term dependencies due to issues like vanishing and exploding gradients, which prevented scalability to large datasets and complex tasks [27].

Reference [28] proposed long short-term memory networks (LSTMs) to address the shortcomings of RNNs. LSTMs introduce a gating mechanism for input, output, and forget gates that allows the network to learn how to preserve and control information over long sequences. This helped LSTMs excel in tasks that required long-range context, such as machine translation or speech recognition. Despite their potential, LSTMs were too computationally costly and could hardly be parallelized, which limited scalability.

The introduction of transformer architectures marked a significant leap forward in natural language processing. Authors in [29] proposed the transformer model in their paper, “Attention Is All You Need,” which replaced recurrent structures with self-attention mechanisms. This architecture could process all sequences in parallel, keeping track of long-range dependencies. Transformers allow much easier parallelization and are especially suited for large-scale language modeling.

These early models, including RNNs, LSTMs, and transformers, were the backbone of progress in natural language processing. While RNNs and LSTMs overcame the problem of sequential data processing, transformers introduced scalable architectures that underpin modern LLMs, such as GPT and BERT. Each innovation contributed essential building blocks that collectively allowed for the creation of highly effective and versatile language models.

2) GENERATIONS OF LLMs

At the very center of most LLMs lies a transformer architecture, a model reliant on self-attention mechanisms to efficiently represent long-range dependencies of text. Popular examples include the GPT series from OpenAI, Google’s BERT, and the PaLM models. All have shown outstanding results in a wide array of natural language processing tasks, such as translation, summarization, question-answering, and many more. LLMs have evolved, for the most part, through various milestones. Each milestone introduced innovative techniques for improving the natural language processing capability of LLMs. Over time, the number of parameters in LLMs has increased, leading to greater model complexity. Table 2 reports a comparison of different LLMs in terms of input, advantages, and disadvantages. This comparison becomes important as the main core model for LLM-based energy systems is usually these models.

Recent evaluations have shifted from simply checking accuracy to running *stress tests* that target specific capabilities. For example, SoEval [30] checks how well models keep structured formats like JSON or XML; Q-Bench+ [31] measures basic visual perception and pairwise comparisons; DEF-MT [32] evaluates multi-turn task planning and completion; and LIBRO [33] tests software engineering skills

by reproducing bugs at scale. Shorter psychometric-style benchmarks show that even 60 carefully designed questions can provide reliable results for comparing humans and AI [34]. At the same time, system-level studies combine quality with factors like speed, latency, and memory use across different hardware to reflect real deployment constraints [35].

Instead of relying only on static leaderboards, newer evaluations focus on five main areas: (i) *hallucination and factual accuracy*, tested under noisy data and image–text grounding, using methods like contrastive training and targeted instruction data [21], [22], [23]; (ii) *retrieval-based reasoning*, where models pull in external knowledge during decoding (e.g., Retrieval-In-Decoder) rather than just from context [24]; (iii) *system-level performance* for real-world assistants, checking things like truthfulness, bias, toxicity, contextual precision/recall, and knowledge retention, often in RAG-based build-and-test pipelines [49]; (iv) *stability*, by measuring how repeatable fine-tuning is across multiple runs (e.g., QLoRA variance) and how it affects reported improvements [50]; and (v) *efficiency*, looking at trade-offs when using quantization or PEFT to save memory and increase speed while keeping accuracy intact [51], [52].

C. TECHNIQUES, TRAINING, AND FINE-TUNING PARADIGMS

1) PRETRAINING ON LARGE CORPORA

LLMs are pre-trained on large and diverse datasets designed to provide comprehensive knowledge about linguistic patterns, domain-specific knowledge, and cultural contexts. Examples of such datasets include Common Crawl [53], a repository of web pages that provides broad coverage of current language usage in many domains, and BooksCorpus [54], a dataset offering rich and coherent text from published books, suitable for capturing narrative structures and long-form dependencies.

Other datasets play equally critical roles in expanding the capabilities of LLMs. For instance, Wikipedia [55] offers an important complement of structured encyclopedic content; ArXiv [56] and PubMed [57] provide significant scientific and technical insight. The Pile [58], a 22-component dataset, integrates high-quality sources such as PubMed, GitHub, and Stack Exchange, offering diverse linguistic features from formal technical documents to informal conversational threads.

Multilingual datasets such as CC100 [59] further extend the linguistic reach of LLMs by including text in more than 100 languages, allowing for effective cross-lingual generalization. GitHub repositories [60] provide programming language data, which is especially important in LLMs that are focused on code generation and comprehension tasks.

Additionally, conversational datasets like Reddit [61] and OpenAI Conversations [62] provide instances of human speech, where LLMs excel—conversational fluency and an ability to react contextually to whatever was mentioned in a conversation. Finally, instruction-tuned datasets (FLAN [63]

TABLE 2. Comparison of different LLMs.

Model	Input	Advantages	Limitations
GPT-1 [36] (2018)	Text	Generative pretraining significantly boosts NLP performance	Small scale; no content filtering; narrower capabilities
BERT [37] (2019)	Text	Bidirectional context modeling; effective fine-tuning on minimal data	No built-in safety filtering; training-data bias; hallucinated outputs
GPT-2 [38] (2019)	Text	Large-scale pretraining; good multi-domain dialogue	Repetitive outputs; lacks reasoning; needs domain-specific tuning
GPT-3 [39] (2020)	Text	Fluent text; coherent large-scale generation	Lacks robust semantic understanding; plausible but wrong answers
GPT-4 [40] (2023)	Text + Images	Multimodal understanding (text + images)	May misinterpret images; occasional hallucinations
PaLM 2 [41] (2023)	Text	Efficient; better multilingual, coding, and reasoning skills	Still hallucinates; needs HPC; experimental advanced features
Gemini [42] (2023)	Text, Images, Audio, Video	Strong multimodal reasoning; expert-level performance; efficient	Requires large compute; may produce biased/hallucinated responses
BLOOM [43] (2023)	Text	Open-source; multilingual; transparent development	Resource-heavy; training data bias; hallucinations possible
Arctic-TILT [44] (2024)	Text + Images	Strong document understanding; efficient multimodal fusion	Training data bias; hallucinations remain
LLaMA-VID [45] (2024)	Text + Images	Token-efficient video processing; handles long video content	Hallucinations; needs task-specific training
GPT-4o [46] (2024)	Text, Audio, Image, Video	Fast, multimodal; excellent multilingual support	Risk of sensitive inferences; accent issues; hallucinations
Llama 3 [47] (2024)	Text	Very large (405B); multilingual; strong reasoning	Possible hallucinations; early multimodal capability
DeepSeek-V3 [48] (2024)	Text	671B MoE; efficient FP8 training; near SOTA	Factual errors; MoE complexity; HPC needs; bias risk

and P3, Public Pool of Prompts [64]) are created to enhance task-following and generalization capability.

In practice, modern systems increasingly *augment* pre-training with retrieval to improve grounding and factuality. Recent work embeds retrieval *within* the decoder to directly guide next-token predictions (RID), enabling explainable, document-linked reasoning. This is further supported by engineering guidelines for deploying production-ready RAG assistants and domain-specific RAG systems [24], [49], [65].

Beyond corpus-scale pretraining, retrieval is now applied not only as external context but also *inside* decoding to bias generation toward grounded evidence. Retrieval-In-Decoder illustrates this integration and is commonly evaluated on explainable QA tasks and citation faithfulness [24].

2) FINE-TUNING STRATEGIES

a: TASK-SPECIFIC FINE-TUNING

This approach focuses on adapting the model to a particular task, such as sentiment analysis, question answering, or named entity recognition, by training it on a labeled dataset tailored to that task. Recent methods adapt decoder-only LLMs to work like discriminative encoders by adding bidirectional attention and soft-prompt encoders. This approach achieves state-of-the-art results on multi-dataset text classification while keeping the efficiency benefits of LoRA [66].

b: DOMAIN-SPECIFIC FINE-TUNING

Domain adaptation aims to adapt LLMs to specific domains like medical, legal, or energy text. This is typically achieved

through continued pretraining on domain-specific data and supervised fine-tuning. For example, BioBERT [67] was fine-tuned on biomedical texts to improve performance on domain-specific applications such as PubMed abstracts. Domain adaptation often involves Continued Pretraining and Supervised Fine-Tuning. In safety-critical or noisy domains, using instruction datasets tailored to a model's specific *hallucination profile* reduces cross-modal errors more effectively than generic instructions, leading to more reliable and grounded answers on LVLM benchmarks [23].

c: EFFICIENT FINE-TUNING

This category focuses on methods that reduce the computational cost and memory requirements of fine-tuning large models. These methods allow for efficient adaptation without updating all model parameters. Key strategies under this category include:

- *Adapter-Based Fine-Tuning*: Introduces lightweight, task-specific modules into the pre-trained model, leaving the main parameters frozen. This reduces memory usage and computational costs while maintaining flexibility for task adaptation [68].
- *Low-Rank Adaptation (LoRA)*: Freezes the pre-trained model parameters and introduces low-rank matrices that can be trained efficiently. LoRA significantly reduces the computational burden while maintaining model performance [69].

Beyond adapters and LoRA, selective-parameter PEFT methods (e.g., BitNorm/BitMix) update only about 0.2% of the model's parameters, with no extra cost at inference, yet

still match or even outperform full fine-tuning on language and multimodal tasks [51]. In addition, combining 4-bit quantization with PEFT gives large memory and speed improvements for models like LLaMA, LLaMA-2, and GPT, with only small accuracy drops—making them practical for edge devices or single-GPU setups [52]. Finally, studies show that QLoRA fine-tuning can vary noticeably across different runs, even with the same settings, highlighting the need for multi-trial reporting and confidence intervals in future work [50].

d: PROMPT-TUNING AND PREFIX-TUNING

These methods focus on optimizing the model's input representation rather than fine-tuning the model's parameters. Task-specific embeddings (soft prompts) are prepended to the input tokens, and only these embeddings are optimized during training. This makes the method computationally efficient and effective for simpler tasks like classification. Prefix-tuning further enhances this approach by inserting learnable embeddings at each transformer layer, providing finer control over the model's attention mechanisms, which is particularly useful for sequence generation tasks [70], [71].

Concurrently, training-time objectives that reshape the representation space with hallucination-aware negatives (e.g., HACL) reduce modality gaps and lower error rates on multimodal hallucination tests, complementing prompt/prefix tuning for robust generation [21].

e: INSTRUCTION AND REINFORCEMENT LEARNING FINE-TUNING

This category includes methods that enhance the model's ability to follow natural language instructions or align with user preferences through reinforcement learning. Instruction fine-tuning involves training models on datasets that contain explicit instructions or guidelines for performing tasks, improving their generalization across tasks. Reinforcement learning fine-tuning, such as in InstructGPT [62], combines supervised fine-tuning, reward modeling, and human feedback to optimize model behavior based on user goals and preferences. Instruction tuning is becoming more *targeted*: by identifying model-specific weaknesses and creating balanced positive/negative training examples, researchers can reduce hallucinations without costly after-the-fact fixes, especially for LVLMs [23]. In addition, contrastive decoding methods designed for noisy retrieval help improve factual accuracy during inference [22]. Reporting practices are also improving: beyond accuracy and robustness, recent work recommends publishing variability across multiple PEFT/QLoRA runs and including hardware-related measures (throughput, latency, memory use) so results are both reproducible and practical for deployment [35], [50].

3) ADVANCED LLM TECHNIQUES IN ENERGY SYSTEMS

While various pre-trained LLMs can be adapted for energy systems, certain techniques significantly enhance their

capabilities for specialized tasks. These techniques can be categorized as Advanced LLM Methods and are widely applicable across domains, including energy systems, to solve complex challenges in decision-making, data retrieval, and real-time task execution.

a: CHAIN-OF-THOUGHT (CoT)

CoT prompting helps LLMs reason through complex tasks by generating intermediate steps. In energy systems, it enhances tasks like maintenance prediction and decision-making. For instance, a hierarchical CoT with scoring ensures logical, accurate reasoning, improving knowledge extraction from diverse energy data [72].

b: RETRIEVAL-AUGMENTED GENERATION (RAG)

RAG combines the power of LLMs with external knowledge sources by augmenting the model's generation ability with real-time data retrieval. In energy systems, RAG can be used to provide real-time information from databases, documents, or IoT sensors, enhancing the model's ability to generate accurate responses or predictions based on the latest available data [73].

c: LLM AGENTS

LLM agents are autonomous models that interact with their environment and take actions based on the knowledge they acquire. These agents are ideal for automation in energy systems, such as predictive maintenance, real-time load balancing, and dynamic grid optimization, where decision-making can be automated based on LLM capabilities [74], [75].

D. ENERGY-SPECIFIC LLM-BASED MODELS

LLMs have gained significant traction in various domains, including energy systems, where they are leveraged for forecasting, optimization, and improving operational efficiency. The application of LLMs in energy systems spans several tasks, such as energy demand forecasting, renewable energy generation prediction, power system optimization, and building energy modeling. These models help address complex, large-scale problems by extracting meaningful insights from diverse data sources, enhancing decision-making processes.

1) ENERGY LLM MODELS AND APPLICATIONS

Table 3 summarizes several prominent energy system models based on LLMs, highlighting their year of introduction, base model, training data, and the specific applications they address.

Figure 3, extracted from Table 3, is a timeline of LLM use in energy systems, visualizing their integration into tasks from 2022 through projected developments in 2025. Each box represents a project or model, and the color corresponds to the specific LLM base model used (e.g., GPT-4, LLaMA, Mistral-7B, etc.). Key insights include:

TABLE 3. Energy system models based on large language models.

Model Name	Year	Base Model	Training Data	Application
DT-GLLM [76]	2024	GPT-4	18 and 85 legal ordinance documents	Wind energy siting ordinance extraction
EF-LLM [77]	2024	Transformer-based	Multimodal PV, wind, load, and price data	Energy forecasting, sparse data handling, hallucination detection
LLM-SUC [74]	2025	GPT-4 (Autogen)	Wind/load forecast and history	Stochastic unit commitment optimization
Trans-T2V [78]	2022	Autoformer + Time2Vec	China electricity consumption (2009–2020)	Monthly electricity consumption forecasting
Conformer-RLpatching [79]	2024	Transformer + DDPG	SG-126 simulator, 106,820 days	Long-term renewable forecasting with confidence
SPG-LLM [80]	2025	GPT-4o, LLaMA-3.1	OPF data, 65k examples	Solving OPF with graph/tabular input
GAIA [81]	2025	LLaMA2	Simulation and knowledge-based QA data	Dispatch operation, black-start, monitoring
DALINE-GPT [82]	2024	GPT-4o with RAG	34 unseen DALINE tasks	Power system simulation via tool integration
LLM4OPF etc. [83]	2023	GPT-4/GPT-4V	Simulated and technical docs (e.g., FERC)	OPF, EV scheduling, document QA, multimodal awareness
EnergyPlus GPT [84]	2024	GPT-4 + agents	BEM specs and objects via RAG	Building energy modeling and simulation automation
RE-LLaMA [85]	2024	LLaMA 3.1 + LoRA	1450 hydrogen/renewable papers	Research support and domain-specific decision-making
DrAgent [86]	2025	Gemini 1.5 + ReWOO	COMTRADE faults + manuals + RAG	Grid fault analysis via agentic LLMs
aLLarMa [87]	2025	Gemma2, Qwen2.5 + LangGraph	EMS logs, PSS/E test network	Real-time alarm handling and KG reasoning
RePower [88]	2025	GPT-3.5-turbo	Lab + simulation data + UCI datasets	Assistant for parameter prediction and state estimation
eFedLLM [89]	2024	BERT + Federated Learning	Synthetic/theoretical data	Efficient federated inference with memory/SVD tuning
LFLLM [90]	2025	LLaMA 2-7B + PEFT	AEMO, EWELD, REFIT, Norway loads	Generalized STLF across voltage levels
LLM4Dist Reconfig [91]	2025	LLaMA 2	IEEE + UK Power synthetic data	Real-time reconfiguration for loss minimization
CMLLM [6]	2025	LLaMA, GPT, BERT	SCADA data from wind farms	Wind power forecasting via time-series to text
LLM-HEMS [92]	2025	Mistral-7B + ReAct	Simulated HEMS user input	Interactive energy management for end-users
LLM-Power Frameworks [93]	2024	GPT, Claude, Bard	UC test cases + dispatch tasks	UC optimization and novel dispatch methods

Model Abbreviations: DT-GLLM: Decision Tree–Guided LLM, EF-LLM: Energy Forecasting LLM, LLM-SUC: Stochastic Unit Commitment LLM, SPG-LLM: SafePowerGraph, GAIA: Grid AI Assistant, DALINE-GPT: DALINE Simulation with GPT, LLM4OPF/EV/Doc/SA: LLMs for OPF/EV/QA/Situation Awareness, RE-LLaMA: Renewable Energy LLaMA, DrAgent: Disturbance Reasoning Agent, aLLarMa: Alarm-Aware LLM, RePower: Autonomous Assistant, LFLLM: Load Forecasting LLM, CMLLM: Cross-Modal LLM, LLM-HEMS: Home Energy Management System LLM.

- *Early-stage adoption (2022–2023)* focused on foundational tasks such as electricity consumption forecasting (Trans-T2V), EV scheduling (LLM4EV), and document summarization (LLM4Doc).
- *Mid-phase (2023–2024)* introduced models targeting multimodal reasoning (LLM4SA), renewable energy forecasting (RE-LLaMA), and federated inference (eFedLLM), indicating growing sophistication and domain specialization.
- *Recent and future developments (2024–2025)* showcase advanced systems for real-time decision-making (RePower, aLarMa), optimization (LLM-SUC, SPG-LLM), and intelligent automation in power operations (LLM-HEMS, DALINE-GPT).

Overall, the figure illustrates how LLMs are revolutionizing energy systems by enabling more adaptive, scalable, and

intelligent tools across various layers—from generation and planning to monitoring and fault diagnosis.

2) PERFORMANCE BENCHMARKS OF ENERGY LLMs

Quantitatively, LLM accuracy and success rates are now measured across multiple energy domains. DALINE-GPT achieves 96.07% exact code pass and 96.24% tool-run success on 34 simulator tasks [82]. EPlus-LLM reaches 100% correctness in creating valid EnergyPlus building models [94]. Grid-Agent reports violation-resolution success of 100% with Gemini-2.5-Flash and GPT-4.1-mini, 81% with GPT-4.1 and Gemini-2.5-Pro, 63% with Gemini-2.5-Flash-Lite, and 44% with GPT-4.1-Nano [95].

Latency and efficiency show major improvements. EPlus-LLM reduces building energy model (BEM) authoring time from 35–56 minutes manually to about 1 minute automati-

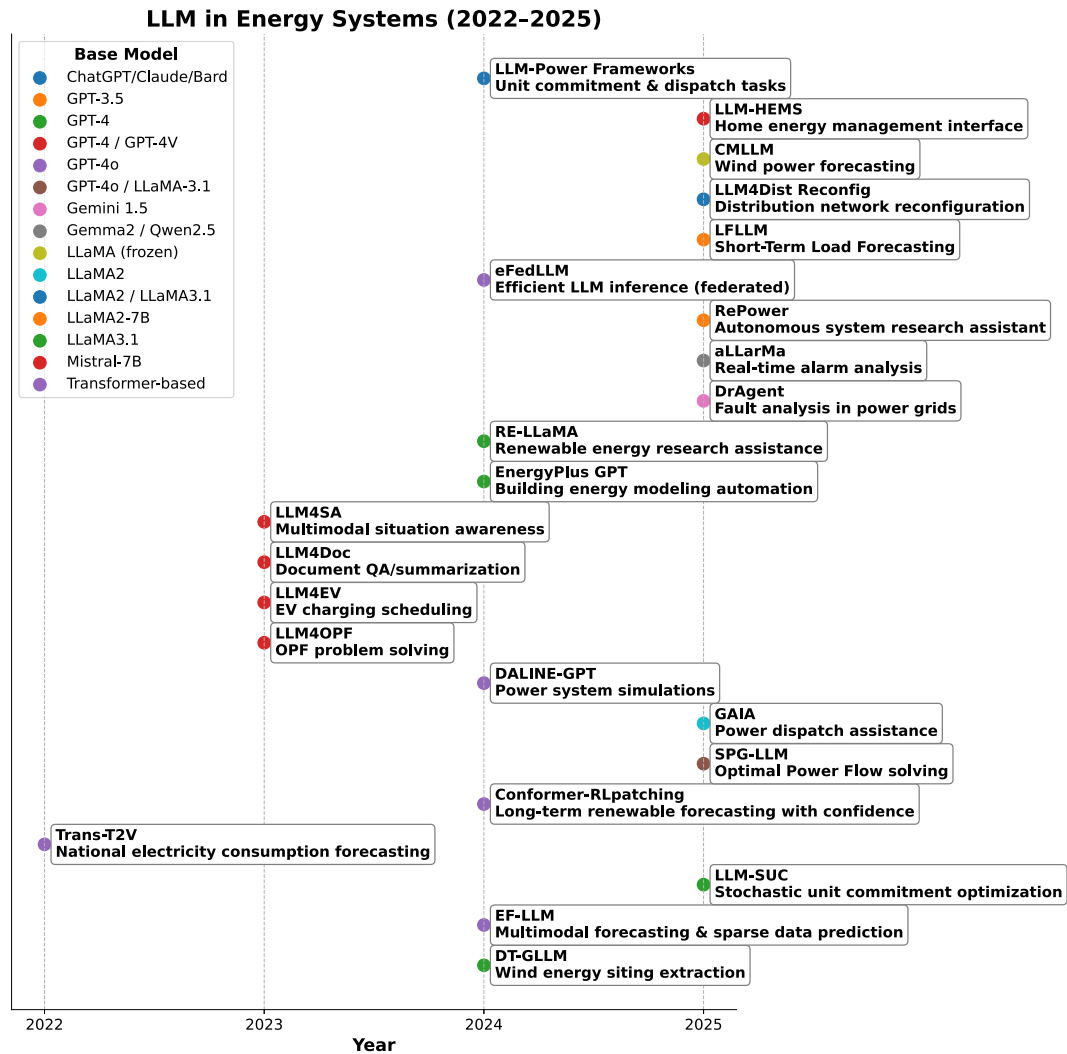


FIGURE 3. Timeline and taxonomy of LLM applications in energy systems (2022–2025), categorized by task and base model. The figure highlights the rapid growth and diversification of LLM-based solutions across various domains such as power system simulations, energy forecasting, load management, and automation.

cally, a reduction of over 95% [94]. DrAgent’s tool caching speeds up repeated tool calls from 0.5 s to 0.0001 s ($\sim 5000\times$) [86]. In Grid-Agent, Gemini-2.5-Flash runs in under 6 s, while GPT-4.1 and Gemini-2.5-Pro take over 16 s [95].

Robustness and safety are also quantifiable. SPG-LLM reduces invalid outputs from 10.8% to 1.3% and improves OPF MSE from 0.113 to 0.020 through instruction tuning and fine-tuning [80]. GAIA achieves the top ElecBench safety score (e.g., dispatch safety 9.750 for GAIA-70B), outperforming general LLMs [81]. aLLarMa lowers the information-loss score by 69% and increases solving and answering accuracy by 8.6% and 12.6%, respectively [87].

Finally, end-task accuracy gains are evident in estimation tasks. RePower reduces average error by 29.07% (up to 39.78%) across parameter prediction and state estimation benchmarks [88]. Together, these results provide concrete, cross-domain evidence of LLM progress in accuracy, latency, and robustness.

E. PRACTICAL TRADE-OFFS FOR REAL-TIME AND EDGE DEPLOYMENT OF LLMs

When LLMs are deployed in energy-constrained environments (such as smart meters, DER controllers, or microgrid EMS), their performance must be balanced with energy use, latency, and hardware limits. The following subsections summarize recent findings and design practices from edge-LLM and energy-systems research, giving a realistic view of how such systems can be deployed in practice [96], [97], [98].

1) ENERGY BUDGETING AND MEASUREMENT

Running models locally improves privacy and resilience but increases the device’s own energy consumption compared to fetching results from remote servers. Studies show that local inference can add significant energy overhead for clients [98], [99]. For grid devices with very small power budgets (such as meters under one watt), key strategies include (i)

low-bit quantization, (ii) batching or windowing, and (iii) reducing FLOPs during decoding. Approaches like AWQ (weight-only) and Atom (weight+activation with KV-cache quantization) cut both energy use and memory while keeping accuracy acceptable [100], [101]. Further savings come from runtime optimizations such as custom kernels, cache-aware scheduling, and operator fusion [102]. To make trade-offs clear, we recommend reporting energy per token (J/token or Wh/response) alongside accuracy [98].

2) LATENCY, QUALITY, AND PLACEMENT

Many protection, control, or user-facing tasks (such as setpoint advice or anomaly triage) require strict latency bounds (p50/p95), which standard server-grade LLMs cannot meet at the edge without compression or algorithmic changes. Split-execution and offloading policies that account for network conditions and device status (battery, thermal limits) can satisfy service-level requirements while saving energy [103]. Speculative decoding (draft-and-verify) shortens response time and reduces device energy use, especially on resource-limited CPUs and NPUs [104]. For interactive devices such as thermostats or HMIs, small on-device models (SLMs) provide fast and power-efficient responses acceptable to human users [105].

3) HARDWARE AND MEMORY LIMITS

Edge devices such as smart meters, gateways, or embedded controllers usually have only 0.5–8 GB of RAM and limited compute capacity. Techniques like unified compression across layers and adaptive layer tuning fit LLMs into these small footprints, often reducing memory use by 2–4× and speeding up inference [106]. As shown in earlier work on quantized neural networks, reducing bit-width can maintain performance while cutting energy and silicon cost [107]. Surveys summarize how design choices—precision, cache use, and scheduling—determine the maximum model size that edge hardware can handle [96], [97].

4) DESIGN PATTERNS FOR METERS AND MICROGRIDS

For smart meters and AMI edges, the focus is on privacy-preserving and communication-efficient methods such as federated or split learning, with small models and selective offloading [108]. Local inference should be preferred for routing-sensitive or private tasks, while cloud offloading can be used for retrieval-heavy or long-context queries if power and network conditions allow [103]. For microgrid EMS and user-side controllers, edge computing can support real-time monitoring and optimization at lower cost and latency. Near-device analytics reduce response times and save backhaul energy [109], [110]. A practical setup combines small local models (for fast parsing and triage) with cloud verification or physics-based simulators for safety-critical tasks. For IoE gateways that connect diverse DERs and sensors, a continuum of edge–cloud processing is recommended: (i) handle routine tasks with compressed

models at the edge, (ii) offload complex reasoning to the cloud selectively, and (iii) cache retrieval results to cut energy and latency [103], [111].

5) REPORTING CHECKLIST FOR DEPLOYMENT

To allow fair comparisons and reproducible results under real-world constraints, we recommend reporting: (1) energy use per token or per response, along with device power limits; (2) latency percentiles (p50, p95, p99) and throughput in tokens per second; (3) peak memory use and model precision (e.g., W4A8, weight-only INT3/4); (4) device type (e.g., meter, gateway, Jetson, Raspberry Pi) with CPU/GPU/NPU details; (5) offloading policy and network assumptions; (6) safety envelope, including guardrails and verification for actions. Following these practices helps align model capabilities with cost, making LLM deployment more practical in meters, microgrids, and substation-adjacent devices [96], [97], [98].

F. FRAMEWORK OF LLM APPLICATIONS IN ENERGY SYSTEMS

To effectively leverage LLMs in energy systems, a structured framework is necessary. Figure 4 illustrates a comprehensive framework for applying LLMs in energy systems, structured across five interconnected stages. This framework involves several key steps, from data collection to decision-making, ensuring that LLMs can be applied to various energy applications such as power systems, building energy systems, and energy forecasting.

1) INPUT DATA

LLMs in energy systems typically work with large volumes of data, including time-series data (e.g., energy demand, production, grid performance), sensor data (e.g., IoT devices in smart grids), external factors (e.g., weather, temperature), historical data, and textual data (e.g., reports, manuals, logs). This data is collected from various sources within the energy system and can be heterogeneous.

2) DATA PREPROCESSING

Before training, raw data must be preprocessed to ensure compatibility with the LLM. This involves data cleaning (e.g., handling missing values or erroneous entries), normalization (for numerical stability), and formatting (e.g., converting time-series or tabular data into text prompts or structured inputs). For domain-specific tasks like forecasting or anomaly detection, additional feature engineering or temporal segmentation may be applied. Tokenization and encoding are also critical steps in preparing the input for the model.

3) MODEL TRAINING AND FINE-TUNING

LLMs are pre-trained on general text corpora but can then be fine-tuned with domain-specific data related to energy systems. Fine-tuning can involve techniques such as

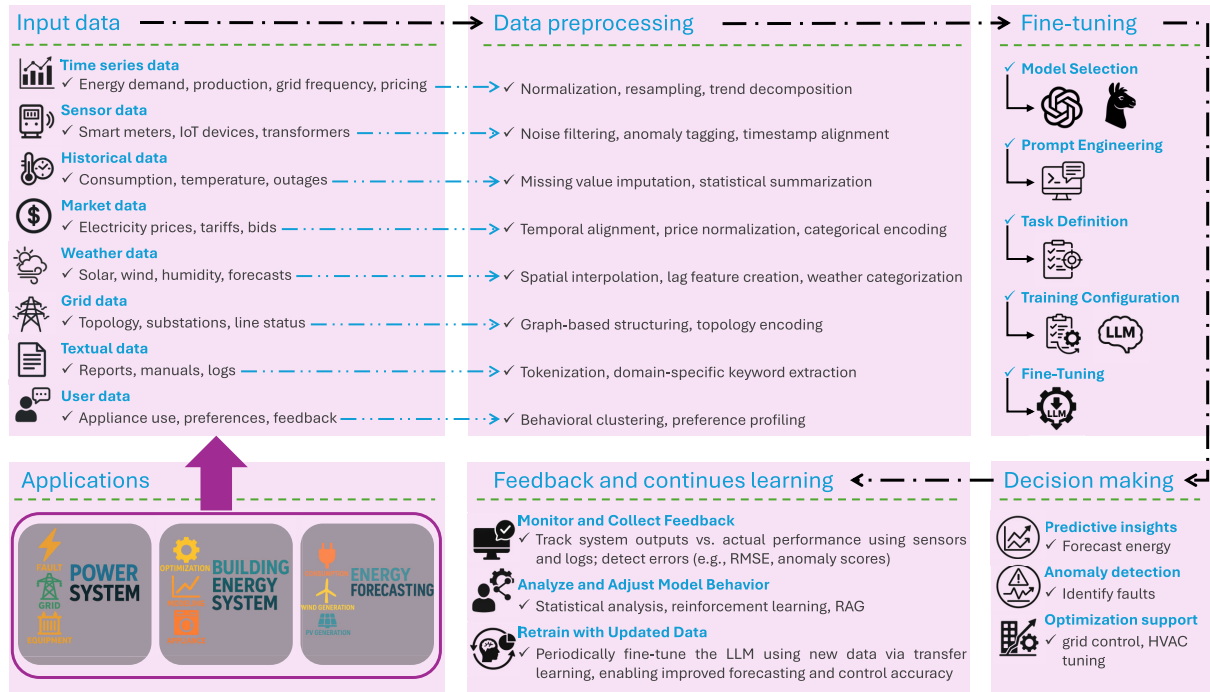


FIGURE 4. Overall framework of LLM in energy systems.

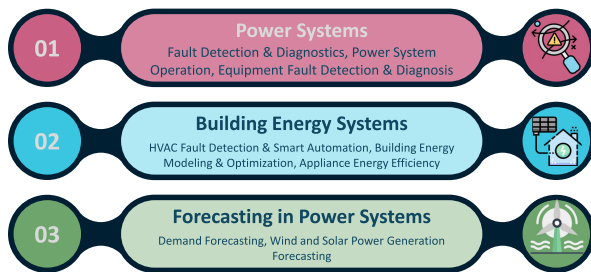


FIGURE 5. Application of LLMs in different areas of energy systems.

task-specific fine-tuning, domain adaptation (e.g., training on energy consumption patterns), and prompt engineering. These fine-tuned models can then be used for energy-specific tasks, such as load forecasting, optimization of power distribution, or failure prediction.

4) OUTPUT GENERATION AND DECISION-MAKING

The fine-tuned LLM processes the input data and generates outputs such as predictive insights, anomaly detection, or optimization recommendations. These outputs are typically used to inform decisions, such as adjusting energy generation and distribution, scheduling maintenance, or optimizing grid performance. The decision-making process often involves integrating the LLM's output with other control systems and models within the energy infrastructure.

5) FEEDBACK AND CONTINUOUS LEARNING

To ensure adaptability in evolving energy systems, LLMs can be enhanced through feedback and continuous learning mechanisms. One approach is Retrieval-Augmented Gen-

eration (RAG), which allows the model to retrieve up-to-date or domain-specific data from external sources—such as real-time SCADA logs or updated market trends—during inference, reducing the need for frequent retraining. Additionally, techniques like online learning, experience replay, and Reinforcement Learning from Human Feedback (RLHF) can be employed to adapt the model incrementally based on new outcomes. Active learning can further guide model updates by focusing on uncertain or high-value examples.

IV. APPLICATIONS OF LLMs IN ENERGY SYSTEMS

LLMs can be applied in energy systems by establishing an interactive platform that facilitates user engagement. Figure 5 summarizes each section and categories investigated in the following. Additionally, Table 4 provides readers with a rapid overview by comparing several references in terms of the model, application, fine-tuning technique, modality, and dataset. Table 5 provides a cross-method synthesis of LLM applications in energy systems, showing representative studies, approaches, data modalities, shared techniques, and limitations across domains such as fault diagnostics, forecasting, markets, and building energy. The comparison highlights both overlaps in methods (e.g., retrieval, prompt templates, PEFT) and critical gaps, including limited benchmarks, weak calibration reporting, and challenges in generalization and robustness.

A. POWER SYSTEM

1) FAULT DETECTION AND DIAGNOSTICS

Fault detection and diagnosis are crucial in power systems to help the grid recover faster, thereby making the grid more

TABLE 4. Overview of papers on large models for forecasting.

Ref.	Model	Application	Fine-tuning	Modalities	Dataset
[112]	BERT4ST	Wind Forecasting	LoRA	Signal, Spatial Data	Chuzhou, NorthSea
[94]	T5	Building Energy Modeling	Supervised Fine-tuning + Prompt Tuning	Text	EnergyPlus Dataset
[113]	GPT-3.5	HVAC Fault Diagnosis	Supervised Fine-tuning	Signal, Text	RP-1312, LBNL, VAV, Chiller Plant
[114]	GPT-based	Energy Optimization	Supervised Fine-tuning + RAG	Text	Energy and Technical Reports
[76]	GPT-4	Energy Policy	Structured Prompting + RAG	Text	US Wind Ordinance Database
[115]	BART, BigBird, Pegasus	Load Forecasting	Supervised Fine-tuning	Text, Signal	Melbourne CBD Smart Meter Data
[116]	GPT-3.5-Turbo	Building Energy Management	Prompt Engineering	Text	ASHRAE Fundamentals Handbook
[117]	TSLLM	Zero-shot Load Forecasting	Prompt Engineering	Text	Australian Solar Household Load Dataset
[118]	GPT-4	Home Automation	Prompt Engineering	Text + Sensor Data	Appliances Power Consumption (Kaggle)

reliable [144]. The work in [119] investigates the integration of LLMs into fault diagnosis in power systems using GPT-4. With tailored prompt engineering and a dataset that includes real-time sensor data, historical fault logs, and component descriptions, the study demonstrates enhanced accuracy, explainability, and coherence in the responses of LLM-based methods. Similarly, [120] explores the application of open-source LLMs, such as Mistral 7B and Llama 3.1, to detect and explain faults in cyber-physical systems. The study evaluates LLMs across two datasets (battery management systems and powertrain systems) and highlights the significance of prompt design and rule inference in improving fault detection accuracy. The paper [145] examines the use of LLMs combined with graph-based approaches for power distribution network fault diagnosis and causality analysis. By utilizing GNNs and RAG, the model enhances fault attribution tasks through dynamic sampling of critical nodes in the power grid network. Experimental results on real power grid datasets indicate that the hybrid LLM-GNN framework performs better than traditional machine learning and deep learning-based methods for fault detection, achieving improved fault localizability of 27% and a reduction of false positives on real systems under difficult grid conditions. Although LLMs show promise, the GPT-4-based diagnostic method still struggles with the complexity of power-grid data and remains a black-box model; the authors themselves note that such models require more domain knowledge and careful prompts to avoid misdiagnosis [119]. Open-source models like Mistral and Llama3.1 rely heavily on prompt design and rule inference, meaning their performance may degrade without domain-specific prompts. Hybrid systems that fuse fuzzy logic with LLMs show improved metrics—precision, recall and F1-score improved over standalone fuzzy or LLM models and false positives/negatives are reduced [123]—but these gains depend on the quality of fuzzy rules

and training data. Compared with graph-based LLM-GNN hybrids, which improved fault localizability by 27% and reduced false positives, the descriptive results indicate better localisation but also reveal that scaling such models across varied grids requires larger datasets and dynamic sampling strategies. Overall, these studies underscore that LLM-based diagnosis should be critically compared with traditional and hybrid methods rather than accepted as inherently superior.

2) POWER SYSTEM OPERATION

Power system operation involves real-time control and management of electric grids to deliver stable, reliable, and efficient generation, transmission, and power supply. The work in [121] explores the potential of LLMs to serve as independent research assistants for power system studies. The authors introduce RePower, an autonomous research platform that allows LLMs to transition from being problem-solving agents to active research designers. However, as noted in [121], general-purpose assistants (e.g., FunSearch or generic GPT-4) can struggle on tasks that require extensive reasoning, feature engineering, or direct interaction with physical equipment, underscoring practical limits in operational settings.

In [82], the application of LLMs in power system simulation is explored to address their limitations in handling new simulation toolboxes. The results demonstrate a significant improvement in GPT-4o's coding accuracy from 0% to 96.07%, which is superior to ChatGPT-4o's regular web interface accuracy of 33.8%, reflecting the potential role of LLMs as intelligent research aids for power system simulation and optimization. Nevertheless, [82] also reports that even state-of-the-art models may fail to construct small distribution networks or to write simple power-flow code without substantial prompt engineering.

TABLE 5. Cross-method synthesis of LLM applications in energy systems: overlaps, differences, shared techniques, and methodological limitations.

Task / Domain	References	Core Approach (Model + Tuning)	Data / Modality	Shared Techniques (Overlap)	Methodological Limitations (Gaps)
Power faults & diagnostics	[119], [120]	GPT-4 / Mistral / Llama; prompt engineering; rules/RAG scaffolding for fault triage	SCADA/event logs; sensor time series; textual fault logs	Template prompts; retrieval of manuals/standards; explanation-first outputs	Few cross-utility benchmarks; weak uncertainty reporting; limited prompt/ablation studies; robustness to noisy logs under-tested
Power system operation & simulation	[82], [121], [122]	Agentic LLMs with tool-use; code synthesis + verification; feedback-driven multi-agent orchestration	Tool APIs (DALINE, MATPOWER), specs, worked examples	Toolformer-style function calling; test-driven self-reflection; example retrieval	External baselines scarce; cross-tool reproducibility; energy/latency of agent loops not documented
Equipment condition & maintenance	[123]–[127]	Fuzzy+LLM fusion; multimodal LLMs (LLaVA-style) fine-tuned; digital twins + DETR variants; LLaMA3-8B with signal prompts	Images; thermal/oil/vibration; transformer/ substation reports	Multimodal prompting; KG/digital-twin context; vision backbones with LLM-generated reports	Domain-internal metrics; few public datasets; safety validation limited; small-sample overfitting risks
Energy markets & policy	[76], [128]–[131]	ChatGPT/RL agents for trading; BERT-BiLSTM sentiment; LLM + decision-tree extraction for legal corpora	Market series/order books; social/news text; zoning/ordinance documents	RAG over policy corpora; sentiment features as exogenous signals; structured IE to machine-readable forms	Causality vs correlation unclear; backtest leakage/overfitting; poor regime-shift analysis
Building HVAC FDD	[113], [132]	GPT-3.5 with PEFT/LoRA; text reasoning over anomalies; few-shot diagnostics	BMS logs, alarms, operator notes	PEFT for small-data; template prompts; explanation and action suggestions	Calibration rarely reported; limited cross-building generalization; sparse human-in-the-loop evaluation
BEM / simulation & optimization	[84], [94], [114], [133]	Multi-agent LLM planners; Plan-T5 fine-tuned to synthesize EnergyPlus files; RAG to standards/specs; NL interfaces to BEM	BIM/specs text; EnergyPlus inputs/outputs; audit reports	RAG grounding; code synthesis + auto-checks; structured-output constraints	Comparisons vs established BEM workflows limited; error propagation to control not quantified
Smart appliances / HEMS	[118], [134], [135]	Mistral/GPT-4 conversational control; ReAct agents for IoT orchestration; rule synthesis for routines	User text; IoT/sensor states; appliance metadata	Action templates; preference grounding; safety/comfort constraints	Latency/energy budgets unreported; guardrails/rollback behavior not evaluated
Load forecasting (STLF)	[115], [117], [136]–[139]	Zero-shot/multi-task LLMs (TSLLM, LFLLM); PEFT adapters; seq2seq T5; NL-to-series reformatting; probabilistic outputs (Chronos)	Smart-meter/feeder series; text+signals	Task-as-text reformulation; PEFT for compute efficiency; prompt templating	Parity vs top TS baselines mixed; context leakage risk; uncertainty calibration seldom audited
Price forecasting	[140], [141]	GPT pipelines with sentiment; agentic bidding + sentiment fusion and market features	Market series; news/social text	Text signals as exogenous features; explainable prompting	Backtesting rigor/rolling evaluation; sensitivity to sentiment sources; regime-shift handling
Wind / PV forecasting	[112], [142], [143]	BERT4ST (LoRA) for spatio-temporal wind; STELLM with spatial/temporal prompts; GPT-Agent for probabilistic PV using weather text	SCADA wind; meteorological text/APIs	Spatial/temporal prompting; weather-KG/RAG; zero-/few-shot setups	Limited cross-site transfer; weak physical constraints; early-stage uncertainty treatment

In related work, the same authors in [122] explore the use of LLMs in power system simulation by circumventing their inherent limitations in reasoning, knowledge retrieval, and management of simulation parameters. The model is evaluated on 69 diverse tasks from DALINE and MATPOWER and achieves success rates of 93.13% and 96.85%, respectively. An in-depth analysis in [122] finds that LLMs lack domain-specific knowledge, have restricted reasoning, and cannot reliably apply simulation functions and parameters, leading to semantic drift and erroneous outputs; consequently, frameworks such as RePower and multi-agent schemes add retrieval-augmented generation, tool learning, and feedback loops to mitigate these issues—at the cost of higher complexity and engineering effort.

3) EQUIPMENT FAULT DETECTION AND DIAGNOSIS

Traditional fault diagnostic methods rely on rule-based systems with the support of an expert or machine learning

that would typically require manual feature engineering and significant human expertise that may be automated to some degree by fuzzy logic. In [123] a new approach to electrical equipment fault diagnosis is presented which integrates Fuzzy Logic Systems with LLMs to enhance decision-making accuracy. It generates an electrical fault dataset, applies fuzzy logic for classification, and integrates LLMs to enhance decision-making using textual data improving accuracy. A comparative analysis shows that the fuzzy-logic–LLM fusion surpasses standalone fuzzy or LLM baselines in precision, recall, and F1 while reducing false positives/negatives; however, its gains depend on the quality of expert-defined rules and the available training data.

Multimodal LLM brings some benefits in grid equipment maintenance. The paper [124] proposes a multimodal domain-specific LLM design named SubstationAI for analyzing faults in substation equipment. An instance-specific dataset of 40000 comprising images, defect tags, and analysis reports was created and used for fine-tuning

the LLaVA1.5-7B model. Relative to general multimodal LLMs (e.g., GPT-4 and MiniGPT-4), SubstationAI reports higher accuracy, clarity, and completeness due to supervised fine-tuning and domain-specific knowledge enhancement, but this comes with the cost of large curated datasets and an external knowledge base, which can limit scalability. In [125], a new approach integrating digital twin and multimodal LLM is proposed for transformer maintenance that overcomes limitations of traditional monitoring. The study [126] proposes a state-of-the-art framework with enhanced deep vision models and digital twin technology for real-time health monitoring and maintenance of transformer equipment using DETR+X model. Digital-twin-based multimodal pipelines demonstrate real-time monitoring and improved small-object detection, outperforming YOLOv8 and Deformable DETR on certain datasets, but they rely on complex data augmentation, high-fidelity twin models, and specialized hardware. Transformer health monitoring is proposed in [127] with different data modalities. The method employs ResNet-based feature extractors for extracting features from oil chromatographic, oil temperature, and core ground current data and projects them into a shared feature space using an input projector. The shared features are utilized as prompts for the LLaMA3-8B to generate in-depth fault analysis reports. Combining temperature, oil chromatography, and ground-current modalities increases diagnosis accuracy from 82.3% (single modality) to 93.6%, yet BLEU improvements are modest and the overall F1 remains moderate, indicating room for further optimization.

4) ENERGY MARKET PARTICIPATION

Power system energy market participation involves the strategic bidding, trading, and scheduling of electricity by generators, consumers, and aggregators for economic and operational optimality and grid stability. Reference [128] introduces a new system that integrates ChatGPT and reinforcement learning to address the complexity of local energy trading. The system facilitates interactive and auto-trading in Local Energy Markets by enabling Distributed Energy Resources to participate in peer-to-peer energy trading. The paper [129] presents a hybrid model based on BERT and Bidirectional LSTM to predict the sentiment polarity of investors and consumers in the energy market. By analyzing text data from online comments of global events in 2020, the paper illustrates the influence of public opinion on the directions of the energy market. The paper [130] uses generative AI, specifically ChatGPT-4, for optimizing the operations of battery energy storage in the Polish energy market. The paper is concerned with using AI to plan charging and discharging cycles to profit from daily price spreads in energy markets.

5) ENERGY SYSTEM DOCUMENTATION AUTOMATION

The energy system, markets, and policies that govern it are rapidly changing. Therefore, it is crucial to have efficient access to and manage policy data. Historically, analyzing

energy policy has only been possible through significant manual labor. For example, the United States Wind and Solar Siting Regulation and Zoning Ordinance Databases recently introduced by the National Renewable Energy Laboratory [131] required approximately 1,500 human labor hours to collect siting regulation data from local zoning codes in the United States. Reference [76] explores how LLMs can automate the extraction of renewable energy siting ordinances from legal documents to support large-scale energy policy research. It introduces a decision tree framework integrated with LLMs to enhance accuracy in processing complex, heterogeneous legal texts, distilling relevant information, and outputting structured, machine-readable data.

6) NEXT STEPS

Key directions for advancing LLM use in power systems include:

- Reproducibility through open benchmarks, anonymized logs, and transparent reporting of prompts and calibration metrics.
- Operational safety with deterministic tools, provenance logs, and human-in-the-loop trials before live deployment.
- Privacy-preserving methods such as federated learning or on-premise fine-tuning for sensitive environments.
- Governance practices including confidence scoring, traceable outputs, and regulatory guidance on transparency and auditability.

These steps provide a practical roadmap for safe and scalable LLM adoption in critical grid applications.

B. BUILDING ENERGY SYSTEM

1) FAULT DETECTION AND DIAGNOSTICS

Heating, ventilation, and air conditioning (HVAC) systems are the largest energy consumers in buildings and are responsible for approximately 40 %–80 % of the total building energy consumption [146]. Fault diagnosis technologies stand out as highly effective energy conservation tools for HVAC systems [147]. From a critical standpoint, LLM-based FDD should be compared against established AI FDD pipelines that already report strong performance yet suffer from data scarcity, domain shift, and sensor noise. Timely removal of HVAC system faults can save approximately 15 %–30 % of the energy consumed by buildings [148]. In [113], they introduce a new method for fine-tuning LLMs for HVAC fault diagnosis. LoRA is used as a fine-tuning technique. Taking the GPT-3.5 model as a baseline, the study shows spectacular performance improvement, with accuracy increasing from 29.5% to nearly 100% on test datasets. However, near-perfect gains reported after fine-tuning can be confounded by narrow benchmarks or prompt leakage; robust comparisons should include cross-building generalization, fault-injection stress tests, sensor dropout/latency scenarios, and cost-sensitive metrics (missed faults vs. false alarms), not accuracy alone. The paper [132]

explains how the use of LLMs, particularly fine-tuned for time-series anomaly detection and fault diagnosis via text, can be used to make HVAC systems more efficient by predicting faults and suggesting maintenance. The study concludes that employing LLMs in HVAC fault detection not only improves predictive maintenance but also saves downtime and loss of energy as well. Current LLM studies often excel at textual fault narratives and operator logs, but are less validated on raw, high-frequency telemetry unless coupled with time-series encoders. Interpretability add-ons (e.g., Shapley + LLM explanations) help operator trust but can be unstable with correlated features, so human-in-the-loop verification and schema-constrained outputs are advised.

2) ENERGY MODELING, SIMULATION, AND OPTIMIZATION

LLMs can be used in building energy modeling, simulation, and optimization, enhancing efficiency and sustainability. In [114] a multi-agent system combined with LLM is proposed to enable the automation of building energy optimization processes. In light of the fact that over 85% of building information is unstructured, the system is designed to support diverse data formats. Experimental evidence indicates that the technique significantly reduces physical effort in carrying out energy audits and improves decision-making effectiveness. The paper [133] presents the use of LLMs in automating metadata normalization in Building Management Systems. For metadata normalization, classical NLP/ML baselines (CRF/MLP active learning, SeNsER, RoBERTa transfer) have reported up to 99% with expert input but as low as 42% without, 67% F1, or >70% accuracy on real-world datasets; direct comparisons remain difficult due to heterogeneous corpora and metrics.

In [94], EPlus-LLM is introduced which can automate the creation of building energy models from natural language descriptions. The method employs a fine-tuned Flan-T5 model to interpret user inputs and produce correct modeling files for EnergyPlus simulation. Methodologically, auto-modeling claims require careful qualification: [94] reports a 100% simulation success across 152 validations, yet the same paper notes that even *minor punctuation or geometric deviations* can break EnergyPlus runs, highlighting brittleness and the need for rigorous validity checks and unit tests on generated IDF files. The article [84] leverages LLMs like GPT-4 combined with EnergyPlus. The study is focused on enriching user interaction using natural language interfaces, generating and editing simulation inputs, and post-processing outputs economically. Natural-language interfaces improve usability but must include guardrails (schema-constrained decoding, physics-based validation) to prevent hallucinated parameters entering safety-critical simulations. Hence, fair evaluation should standardize metrics (valid IDF rate, simulation pass rate, energy KPI error), report prompt sensitivity/ablation, and test cross-climate/building types. In [116], Shapley values and LLMs are integrated to enhance the explainability of machine learning control (MLC) systems for building

energy management. By the capacity of Shapley values to interpret and that of LLMs to reason, the study aims to bridge the trust gap between control operators and algorithm designers.

3) SMART APPLIANCE CONTROL

LLMs are revolutionizing appliance use in buildings through improved energy efficiency and control strategy automation. [134] explores the use of LLMs, specifically Mistral 7B, to optimize the operation of household appliances in order to save energy costs. Critically, LLM scheduling remains preliminary: [134] reports *acceptability* rates around 0.93–0.97 across months with 5–18% missing load schedules; invalid intervals make cost comparisons *not meaningful*, indicating evaluation fragility and the need for constraint-aware decoding and fallback optimization. The paper [118] introduces GreenIFTTT, a GPT-4-powered conversational agent that helps create sustainable home automation routines for residents. It investigates the application of LLMs to simplify appliance usage, optimize energy consumption, and encourage eco-friendly behaviors. GreenIFTTT's assessment emphasizes UX (SUS/UEQ/PSI) over measured kWh or bill savings, so real-world energy benefits remain to be causally demonstrated. In [135] combined AI-powered LLM-ReAct method is used for simplifying energy management in smart buildings. The system integrates LLMs with IoT data in real time to manage appliances, predict energy consumption, and automate energy efficiency interventions. With wireless control of electric appliances and BIM, the framework provides optimized energy conservation and human engagement with smart environments. The ReAct-based agent achieves a high success rate *in virtual simulations* but lacks safety proofs for live control; deployment should add action-filters, device/comfort constraints, and rollback plans before field trials. For a fair comparison with rule-based/MILP controllers, future studies should report standardized metrics (comfort violations, peak-shaving, cost savings under tariffs) and run A/B tests on real homes.

4) NEXT STEPS

Key directions for advancing LLM use in building energy systems include:

- Create open, multi-building corpora with unified fault taxonomies, anonymized BMS alarms, and benchmarks that include calibration metrics (e.g., ECE) and operator-in-the-loop studies.
- Standardize natural-language-to-BEM JSON schemas with strict validators, assess workflows against expert baselines, and quantify how modeling errors propagate into control decisions.
- Evaluate appliance-control agents on real devices with clear latency and energy profiles, add safety guardrails and rollback plans, and enable preference learning under privacy budgets.

- Encourage utilities and building authorities to provide delayed, machine-readable data (BIM/BMS metadata, demand-response hooks) under privacy-preserving conditions to support reproducibility and research.

Together, these steps highlight the need for shared datasets, safety mechanisms, and standardized interfaces to ensure that LLM-based solutions for building energy systems are accurate, transparent, and deployable in practice.

C. FORECASTING IN POWER SYSTEM

1) DEMAND FORECASTING

Current advances have noted LLMs achieving exceptionally high results in zero-shot load prediction through the implementation of multi-task learning models in predicting power load without requiring monumental retraining [117]. [136] introduces LFLLM, a Transformer-based LLM specifically designed for short-term load forecasting at multi-voltage levels in power systems. A Parameter-Efficient Fine-Tuning strategy is employed to reduce computational overhead while maintaining model accuracy. The analysis indicates that LFLLM attains higher accuracy and reliability than transformer baselines but still incurs notable computational overhead and depends on fine-tuning to manage high-frequency noise, which challenges real-time deployment. The research [115] suggests a new paradigm employing language models, such as BART, Bigbird, and Pegasus, to support energy load prediction. However, cross-building tests show that models trained on a single building generalize poorly to unseen buildings, yielding significantly higher RMSE and MAE than task-specific baselines. In [137] the application of Transformer-based generative NLP models, specifically T5, is investigated for multi-horizon energy consumption forecasting. By defining the forecasting task as a sequence-to-sequence problem, the study converts numerical time-series data into text-like representations that allow the models to process complex temporal and contextual dependencies. Results improve when contextual information and seq2seq framing are included—outperforming LSTM and GRU for day-ahead and week-ahead horizons—but degrade without context and require careful prompt design. [138] suggests ways of converting numerical time-series data into natural language format so that LLMs can process the data more effectively. This general LLM-based framework mitigates hallucinations by language-converting time series and improving data quality; without such enhancements, naive LLMs show large RMSE and higher hallucination rates. probabilistic and deterministic load forecasting is done in [139] using the Chronos model. Both data-rich and data-scarce environments are handled well by the model, outperforming traditional machine learning as well as task-specific statistical models, which tend to be based on huge amounts of such data. Even so, Chronos requires substantial training resources, and evidence for applicability beyond power-load data remains limited.

2) ELECTRICITY PRICE FORECASTING

Electricity price forecasting is essential for optimizing energy market operations, considering supply-demand equilibrium, weather conditions, and fuel prices. The study [140] explains the application of GPT models in predicting electricity price trends. Results indicate that optimized GPT models are more precise and sentiment-correlated than conventional methods, including VADER and BERT, and offer a powerful tool for understanding market dynamics and informing decision-making processes. However, fine-tuned GPT models that extract sentiment from energy news, while outperforming sentiment analyzers such as VADER and BERT in accuracy and Matthews correlation coefficient, rely on high-quality, domain-specific text and carefully crafted prompts, which limits generalization across markets.

In another study [141], LLMs are used in electricity price forecasting by adopting bidding behavior analysis and market sentiment evaluation. Case studies demonstrate how the proposed framework significantly improves both spike and price prediction, and therefore offers high-frequency trading markets with a very effective decision tool. That said, the bidding behaviour and market sentiment agent framework—integrating a fine-tuned LLaMa3 model into a spike-enhanced CTSGAN—achieves lower RMSE and MAE than LSTM, GRU, and AEMO baselines and improves spike detection, but fine-tuning the LLaMa3 8B model takes roughly 49 hours (about two orders of magnitude longer than training LSTM/GRU), demanding substantial computational resources. These trade-offs highlight the need to evaluate computational cost, interpretability, and data requirements when adopting LLMs for price forecasting.

3) WIND AND PV GENERATION FORECASTING

Wind and PV generation prediction plays a significant role in balancing power grids through predicting renewable energy output based on weather, trends, and real-time observations. In [112] BERT is used for spatio-temporal wind power prediction. The research presents BERT4ST, an approach that fine-tunes a pre-trained BERT model to learn spatial and temporal trends in wind power data. Empirically, BERT4ST attains lower errors than DLinear, LSTM, Transformer, Informer, and graph neural networks across multiple horizons, but self-attention models can overfit high-frequency noise, and BERT4ST relies on multi-stage training with increased computational complexity.

Reference [142] suggests STELLM, a novel model that fine-tunes a pre-trained LLM to forecast complex wind speed patterns using seasonal-trend decomposition and spatio-temporal prompts. The model utilizes spatial prompts, which identify geographic relationships between wind turbines, and temporal prompts, which identify short-term patterns in historical wind data. STELLM reduces MAE by about 12% versus baselines through seasonal-trend decomposition and spatio-temporal prompts; however, long sequence modeling

and the need for spatial prompts increase memory use and limit generalization to PV forecasting.

The paper [143] examines the application of LLMs in probabilistic PV power forecasting. The paper introduces a GPT-Agent that converts linguistic weather forecasts from sources like the OpenWeather API into numerical vectors that represent cloud coverage probability. It is demonstrated that incorporating LLM-generated weather interpretations greatly enhances forecast accuracy and interpretability, providing valuable insights into how weather impacts PV generation. Concretely, the GPT-Agent plus XGBoost yields the lowest NMAE (0.0477), MSE (0.0087), and RMSE (0.0935) among compared methods, but the approach depends on accurate, timely weather text and sufficient high-quality linguistic data.

4) NEXT STEPS

Future work on LLM-based forecasting in power systems should focus on:

- Adopting safe evaluation methods: use rolling-origin splits for load, wind, and PV; for price forecasting, apply purged or embargo cross-validation; report both normalized and probabilistic metrics.
- Going beyond point forecasts: provide calibrated probabilistic outputs (intervals, quantiles) and check their reliability with conformal prediction or similar methods.
- Improving weather-data integration: clearly document NWP or reanalysis inputs (e.g., ERA5) and measure terrain- and season-based biases before training.
- Testing generalization: include cross-building, feeder, or site hold-outs and run regime-shift stress tests to reflect real-world deployment.
- Reporting efficiency: publish training and inference costs, memory use, and end-to-end latency to assess practical use in real-time operations.

V. LLMs VERSUS CONVENTIONAL AI

Energy applications impose constraints that differ from standard NLP and computer-vision benchmarks. In load forecasting, recent studies show that time-series foundation models and general-purpose approaches often perform only on par with strong task-specific baselines when tested under realistic data conditions [149], [150], [151], [152]. In interactive human-in-the-loop settings, LLM agents can reduce trial-and-error and improve robustness to extreme events with relatively low additional cost [153]. For privacy- and heterogeneity-sensitive solar forecasting, clustered federated learning with explainability has proven both competitive and practical for deployment [154]. Overall, these results highlight important differences between LLMs and conventional AI in terms of data requirements, interpretability, domain adaptation, and deployment feasibility, especially in mission-critical or resource-constrained grids [155], [156]. Table 6 further contrasts the strengths and weaknesses of LLMs and conventional AI across these aspects, showing

where each approach is better suited and where key gaps remain.

A. DATA REQUIREMENTS AND LABELING BURDEN

Studies in both utility-scale and household settings show that LLMs and time-series foundation models do not always outperform tuned baselines; results depend on forecast horizon, granularity, and data coverage [149], [150]. For hierarchical short-term load forecasting, domain-tailored Transformers and LSTMs remain competitive when trained at each grid level with realistic data [151], [152]. By contrast, interactive LLM agents can reduce human trial-and-error and achieve acceptable accuracy with little additional labeling [153]. For PV and solar forecasting, clustered federated learning uses site heterogeneity and preserves privacy while remaining interpretable [154].

B. INTERPRETABILITY, AUDITABILITY, AND OPERATOR TRUST

Reasoning traces from LLMs are not reliable explanations; energy applications require formal XAI methods and auditable workflows [158], [159]. For control and scheduling, interpretable pipelines such as SHAP-enabled data-driven MPC provide feature attributions that operators can validate [160]. Reviews in load forecasting stress the importance of combining attribution with uncertainty quantification to avoid brittle decisions [157], [159]. A practical pattern is to use LLMs for orchestration and conventional models for prediction, with SHAP or ICE outputs archived for compliance [156], [158].

C. DOMAIN ADAPTATION AND TRANSFER ACROSS GRIDS AND ASSETS

Zero- and few-shot prompts can speed up adaptation to new feeders or mixed energy vectors, but careful validation is still necessary [117], [149]. Classical transfer, where TFT or LSTM models are retrained on new feeders, remains reliable when grid topology or tariff structures change [151], [152]. For integrated energy systems, zero-shot prompts from LLMs provide a starting point for feature engineering and scenario generation, after which domain-specific forecasters can be trained cost-effectively [117].

D. DEPLOYMENT FEASIBILITY: EDGE, LATENCY, AND ENERGY

Critical operations require predictable latency and energy use, which often favors embedded conventional models or TinyML [156], [161]. Inference through API-based LLMs adds external dependencies and variable latency, while distilled or on-device LLMs reduce this but still need considerable resources. Quantized LSTM, CNN, or TFT variants run well on edge hardware with stable timing and low power draw [161]. Guidance from DOE and NERC emphasizes staged rollouts, human oversight, and fallback logic for any AI that affects real-time operations [155], [156].

TABLE 6. Contrasting LLMs and conventional AI for energy-system tasks.

Aspect	LLMs (GPT-style, agents, TS FMs)	Conventional AI (RF/CNN/LSTM/Transformer, statistical)
Data requirements & sample efficiency	Benefit from diverse corpora; zero/few-shot transfer and tool-use help when labeled energy data are scarce; gains are <i>not</i> universal vs tuned baselines [149], [150], [153].	Strong with structured, labeled series; tailored architectures/training for power-load hierarchies remain highly competitive at typical utility data scales [151], [152], [157].
Interpretability & auditability	Native chain-of-thought is <i>not</i> a formal explanation; must pair with XAI (e.g., SHAP on inputs or tool traces) and procedural safeguards [158], [159].	Mature XAI toolkits (SHAP, ICE, counterfactuals); interpretable control (e.g., data-driven MPC with SHAP) provides clearer causal narratives and engineering audit trails [158]–[160].
Domain adaptation & transfer	Zero-shot and cross-domain prompts (e.g., integrated energy systems) show promise; few-shot personalization lowers labeling needs [117], [149].	Transfer via pretraining (TFT/LSTM) and hierarchical conditioning works well; explicit re-training often needed when topology or tariffs shift [151], [152].
Robustness to shifts/extremes	Agent workflows can adapt instructions during extremes (storms/typhoons), improving error markedly at small token cost [153].	Robustness via seasonality decomposition, regime switching, and physics-aware features; often more predictable under distribution shift [157].
Privacy & data governance	Tool-augmented LLMs with federated or edge summarization mitigate centralization, but governance is complex [155].	Federated/clustered training with local features is well-aligned with utility privacy and data residency constraints [154].
Deployment feasibility (latency/compute/edge)	Inference can be heavy; distilled/lightweight or API-based execution adds dependencies; care needed for real-time ops [156].	Broad support for embedded/edge (TinyML, quantized LSTM/CNN); energy/latency budgets are predictable for field devices [161].
Safety, reliability & compliance	Require guardrails, tool validation, and operator-in-the-loop for mission-critical actions [155], [156].	Established verification and fail-safe patterns; easier to certify under deterministic envelopes [156].
Typical wins	Rapid knowledge integration, zero/few-shot adaptability, code/automation for data wrangling and workflows [117], [153].	High accuracy/efficiency for well-instrumented tasks (STLF, PV/wind) with clear features; robust, cheap, maintainable at scale [151], [159].
Typical pitfalls	Hallucinations, non-determinism, cost/latency, opaque reasoning without XAI; harder certification [150], [156].	Limited cross-task generalization; re-engineering needed when assets/tariffs/topologies change [157].

E. SAFETY, COMPLIANCE, AND OPERATIONAL READINESS

Regulatory and operational guidelines recommend a human-on-the-loop approach, deterministic guardrails, and explainable outputs before granting autonomous authority [155], [156], [162]. Conventional models benefit from well-established verification practices such as limits, alarms, and conservative fallbacks, which make certification easier [163], [164]. LLM and agent-based systems should initially be limited to advisory roles, with tool use sandboxed and outputs validated against power-system constraints [156].

F. WHEN TO PREFER WHICH (DESIGN HEURISTICS)

Conventional AI is preferable for embedded short-term load forecasting, feeder-level PV prediction, or protection-adjacent analytics where strict latency, energy budgets, and auditability are required [151], [154], [156], [161]. LLMs and time-series foundation models are more suitable for cross-domain prototyping, rapid what-if analysis, knowl-

edge extraction from manuals, and interactive tuning with sparse labels, provided they are paired with explainability tools and human oversight [149], [153], [158]. In practice, mixed setups are common: LLMs handle orchestration, documentation parsing, and hypothesis generation, while conventional models provide fast, stable predictions at the edge. Both are supported by explainability and governance layers [155], [159].

VI. LIMITATIONS, CHALLENGES, AND FUTURE PROSPECTS

Language model creation and deployment challenges are categorized in Figure 6, which presents a structured overview of the primary limitations and challenges in applying LLMs in energy systems. These challenges fall under three primary categories. Data challenges include the lack of access and sharing of effective energy datasets, the need to generate synthetic data, and the use of domain knowledge

through prompt engineering. The model-related challenges highlight the limitations of existing LLMs to be specific to a particular energy application, keeping interpretability intact, reducing hallucinations, and achieving computational efficiency for real-time deployment. Finally, ethical and safety challenges include protecting privacy, cyber-securing systems, and mitigating bias or disinformation. Addressing these interconnected challenges is pivotal to responsibly and effectively leveraging LLMs for future energy systems. To operationalize these insights, Table 7 condenses specific open challenges and associated research opportunities across data, model, and ethical/safety dimensions.

A. DATA-RELATED CHALLENGES

Effective LLM application to energy systems depends on the quality and availability of data. Power grid, building, and forecasting domain-specific data is usually limited or disintegrated, hindering model training and performance. Data-related challenges are described for each domain and scholarly solutions for overcoming them below.

1) SYNTHETIC DATA GENERATION AND AUGMENTATION

To overcome limitations caused by inaccessible, confidential, or insufficient real-world data, researchers have explored synthetic data generation and augmentation techniques:

- *SubstationAI*: Develops a multimodal LLM-based approach to substation equipment fault analysis, through the creation of a 40,000-entry database containing images, defect labels, and analysis reports. The method enhances dataset diversity towards more robust LLM performance in the fault detection application [124].
- *FD-LLM (Fault Diagnosis LLM)*: Presents a novel LLM-based fault diagnosis framework by innovatively extending LLMs to numerical data inputs. Maps vibration signals into text formats and extracts statistical features in both time and frequency domains, realizing efficient processing of time-series sensor data [168].
- *EF-LLM (Energy Forecasting LLM)*: Proposes an energy forecasting LLM that integrates domain knowledge and time-sensitive data for time-series forecasting. It supports both pre-prediction tasks and post-prediction decision-making, enabling effective energy forecasting even in the presence of bad or incomplete past data [77].

2) PROMPT ENGINEERING AND KNOWLEDGE INTEGRATION

Improving the interpretability and accuracy of LLMs in energy systems can be achieved by integrating domain-specific knowledge through tailored prompts:

- *Prompt-Engineered LLMs*: Context-aware prompt engineering is appropriate for power grid operating conditions to allow models like ChatGPT to generate high-granularity insights for fault diagnosis and operational suggestions. This improves diagnostic accuracy and transparency [119].

- *GAIA (Grid Artificial Intelligent Assistant)*: a benchmark for quantifying general-purpose AI assistants, can help address data-oriented problems in energy systems by ensuring LLMs possess significant features such as processing multi-modal data, web-based retrieval, and constraint reasoning. These features are essential in energy applications where models must integrate different sources such as SCADA logs, weather feeds, and regulation data, which in most instances are real-time. GAIA's focus on tool-use and dynamic access to information makes it a valuable filter for the selection of LLMs that will be effective in critical, information-based smart grid systems [81].
- *RAG-OpsMaintGPT*: To overcome the limitations of fragmented, missing, or inaccessible data in intelligent building systems, the RAG-based approach can be employed. The system, named OpsMaintGPT, employs a knowledge graph constructed using building-specific hierarchies (i.e., floors, zones, sensors, and energy standards) and uses RAG to look up relevant triples and examples for the user's query. This context-specific knowledge is then utilized to generate correct SQL statements that allow the LLM to query properly structured databases correctly, even in situations of absent or incomplete data. The process ensures improved understanding and recovery of functioning HVAC, lighting, and energy information, thus making the model robust under practical situations with the absence of high-quality labeled data. The introduction of RAG allows for dynamic versatility over diverse building structures and enhances the overall generalizability and contextual validity of the outputs of the LLM in maintenance and optimization operations [169].
- *Zero-Shot and Few-Shot Learning Techniques*: Explore the effectiveness of zero- and few-shot approaches with instruction-following LLMs, allowing accurate predictions or diagnostics in new scenarios with minimal or no historical data, significantly addressing data sparsity [181].

3) DATA SHARING AND COLLABORATIVE PLATFORMS

Facilitating secure, privacy-preserving data sharing among stakeholders and researchers can significantly mitigate data scarcity and standardization issues:

- *Federated Learning*: Proposes a privacy-preserving knowledge sharing framework to facilitate few-shot building energy prediction based on federated learning, enabling LLM training across decentralized datasets from multiple buildings without compromising individual privacy [165].
- *RePower*: Introduces RePower, an autonomous LLM-driven research platform that uses a reflection-evolution strategy to independently conduct complex research in power systems. It supports human researchers by acquir-

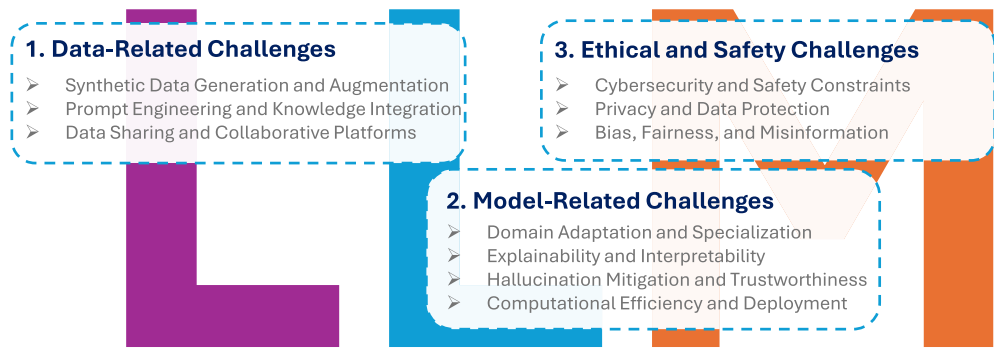


FIGURE 6. Categorization of Key Limitations and Challenges of LLMs in Energy System Applications.

TABLE 7. Open challenges and research opportunities for LLMs in energy systems, organized by category.

Category	Specific Open Challenge	Targeted Research Opportunities	Refs.
Data	Scarcity, fragmentation, and lack of standardization across utilities/buildings	Curated public benchmarks, shared ontologies, and schema alignment; privacy-preserving data collaboratives (federated/consortia); provenance tracking	[81], [165]–[167]
	Synthetic data realism and coverage for rare faults and edge cases	Physics/rules-informed generators; sim-to-real validation protocols; synthetic–real mixture training with shift detection	[124], [168]
	Retrieval quality over heterogeneous sources (SCADA, weather, regulations)	Automated KG construction from hierarchies; RAG with versioned corpora and uncertainty-aware retrieval; dataset documentation practices	[169], [170]
Model	Domain adaptation for safety-critical tasks (power electronics, HVAC control)	Constrained decoding with engineering rules; hybrid LLM+physics/controls; task-specific adapters and prompt-constraint templates	[10], [171]
	Explainability and uncertainty for operational trust	Token/feature attributions aligned to physical variables; counterfactuals under operational limits; calibrated predictive intervals for dispatch/forecast	[172], [173]
	Hallucination and unverifiable recommendations	Verification layers (RAG-to-standards, rule checkers, simulator-in-the-loop); anomaly/hallucination detectors with human-in-the-loop gating	[77], [174]
	Compute/latency constraints at the edge	Quantization/distillation to sub-3B models; split inference (edge/cloud) with strict SLAs; event-driven pipelines for real-time control	[105], [175]
Ethical & Safety	Cybersecurity threats (prompt injection, unsafe control synthesis)	Domain-specific red teaming; IEC 61850-aware intrusion detection; robust prompt hardening and execution guards for control actions	[176], [177]
	Privacy for occupant- and asset-level data	Federated/on-device learning with DP; secure logging and blockchain-backed audit trails; privacy-aware parameter estimation	[165], [178]–[180]
	Bias/fairness and geographic transferability	Bias audits across regions/customer segments; fairness-aware training objectives and reporting; governance checklists for deployment	[145], [173]

- ing data, designing methods, and evolving algorithms to tackle challenging problems [88].
- *Multimodal Data Integration Platforms*: Propose a multi-modal feature learning model based on TimesNet and T2T-ViT for ultra-short-term solar irradiance prediction, integrating diverse data forms such as sensor readings, user feedback, weather, and market signals into standardized and labeled corpora [166].
 - *Blockchain-Based Data Exchange*: Propose an end-to-end blockchain-based solution to provide traceability of energy loss in a distributed network and enhance transaction security, trust, and transparency, ensuring secure, transparent, and tamper-proof data sharing across multiple stakeholders [167].

B. MODEL-RELATED CHALLENGES

Despite their potential, LLMs encounter several model-centric limitations when applied to energy applications. These

issues include high computational demands, hallucinations in domain-specific tasks, and opaque reasoning processes. In the following, we organize the challenges by the solution strategies used to address them.

1) DOMAIN ADAPTATION AND SPECIALIZATION

General-purpose LLMs often lack alignment with the physics-based, safety-critical nature of energy tasks. Domain adaptation techniques — including fine-tuning, model fusion, and prompt engineering — help align models with real-world constraints.

- *PE-GPT*: A variant of GPT specialized for the power electronic domain, PE-GPT, is trained on schematics of power electronics and design constraints. It reduces hallucinations by imposing engineering constraints, such as voltage levels, at inference time, and thereby prevents unsafe or infeasible component suggestions in grid circuits [171].

- *BEM-LM (Building Energy Management LM)*: BEM-LM is adapted to building control tasks by fine-tuning on sensor logs and occupancy schedules. This reduces errors in HVAC scheduling caused by generalized assumptions in vanilla LLMs, thus improving energy efficiency while maintaining comfort [10].

2) EXPLAINABILITY AND INTERPRETABILITY

LLM decision processes are opaque, posing challenges in environments that require traceability and justification. This lack of explainability limits trust and regulatory acceptance, especially in critical infrastructure settings.

- *Explainable AI for LLM*: The approach integrates explainability techniques like LIME, SHAP, and Integrated Gradients to be applied in LLM output analysis, creating feature attribution and transparency in decision-making. The architecture can allow users to make sense of model behavior at the token level and supports both model-specific and model-agnostic interpretation methods [172]. In energy systems, this can be applied to ascertain how LLMs score variables like demand forecasts, weather inputs, or grid conditions when producing their outputs.
- *RAG-AI Assistants*: RAG reduces hallucination in LLM outputs by grounding responses in verified sources such as IEEE standards and OpenEI datasets. In energy systems, this improves explainability and trust, while integration with RDF-based knowledge graphs enhances interoperability across heterogeneous data sources and legacy platforms [170].

3) HALLUCINATION MITIGATION AND TRUSTWORTHINESS

LLMs frequently produce “hallucinated” outputs—responses not grounded in input data or physical laws. In energy systems, such inaccuracies can result in unsafe decisions or mislead forecasts.

- *EF-LLM with Hallucination Detection*: This variant of EF-LLM introduces a hallucination detector that flags outputs diverging from operational norms or historical patterns. For instance, if a forecast suggests an implausible surge in demand, the system suppresses or qualifies the prediction with uncertainty flags [77].
- *PE-GPT with Design Constraints*: By incorporating a rule-based Model Zoo, PE-GPT prevents fabrication in design tasks. For example, it avoids suggesting topologies that violate voltage isolation or current flow conditions, directly addressing hallucination in constrained environments [171].
- *LLM-Based Security Verification Framework*: This framework ensures safe utilization of large language models in power system control operations by verifying generated outputs through RAG and verification against system operation regulations. It cross-references LLM outputs with expert-validated datasets and operational limits to prevent hallucinations and

unsafe control actions. The framework improves trustworthiness through the addition of domain-specific retrieval and verification layers into LLM-based decision pipelines [174].

4) COMPUTATIONAL EFFICIENCY AND DEPLOYMENT

LLMs are large, slow, and power demanding — traits incompatible with real-time or edge-level energy applications. Researchers are now compressing models and distributing inference pipelines.

- *Thoughtful Things (TT)*: To overcome the compute cost challenge of using LLMs in energy-constrained systems, the TT leverages small, sub-3B parameter models (e.g., Phi-2, Gemma) that run fully on-device on a Raspberry Pi 5. By breaking free from the cloud and taking advantage of quantization, the system provides real-time temperature control and user input with minimal compute and power demands—evidence that LLMs can be rendered pragmatic for embedded HVAC applications in energy-constrained environments [105].
- *EDGE-LLM (Edge-Deployed Generative Execution for LLM)*: The EDGE-LLM framework tackles computational complexity by offloading resource-intensive LLM inference to the cloud, while performing lightweight decision-making and latency-sensitive tasks at the edge. This split-execution model enables real-time responsiveness and energy-efficient operation, crucial for applications in smart energy systems where compute power and data privacy are constrained [175].

C. ETHICAL AND SAFETY CHALLENGES

The application of LLMs in energy domains introduces ethical and operational risks. Below, recent solutions are categorized for cybersecurity, safety, privacy, and misinformation.

1) CYBERSECURITY AND SAFETY CONSTRAINTS

LLMs integrated into smart grid and substation environments raise novel vulnerabilities, such as prompt injection, adversarial exploitation, or unsafe control generation. Researchers have begun designing domain-specific defenses and hybrid architectures.

- *ChatGPT for IEC 61850 Cybersecurity*: Explores how ChatGPT can support anomaly detection in smart grid protocols like IEC 61850. The study outlines how human-in-the-loop prompts, aligned with cybersecurity tags, reduce false negatives and improve anomaly traceability [176].
- *Risk Assessment of LLM Integration*: The deployment of LLMs in power systems can introduce security vulnerabilities, including data privacy concerns and operational integrity issues. A comprehensive analysis highlights the need for robust countermeasures to mitigate these risks [177].

- Real substations face protocol-level risks: IEC-61850 GOOSE/MMS traffic on operational networks is usually unencrypted and has been exploited in past attacks. Anomaly detection tuned for IEC-61850 has been tested on real substation traces and can identify attack patterns [182]. This shows that LLM-based monitoring should use packet features linked to real SCADA events, not only toy datasets.
- Control-room practice warns against giving AI full autonomy. Industry standards stress that AI tools should give traceable, procedure-bound recommendations (for example, alarm triage or report drafting), with operators always able to override them [156]. These same rules apply to LLM agents in EMS/SCADA.
- Adversarial risks in state estimation are operational. Black-box attacks on ML-based DSSE have gone undetected and influenced voltage-control logic in simulations, showing how model errors can lead to unsafe controls if LLM/ML outputs are used without checks [183]. New work also shows that CGAN-based attacks can bypass bad-data detection and push state estimation outputs in harmful ways [184].
- For grid-specific LLM threats, prompt injection and knowledge exfiltration have been shown on real smart-grid data. Attackers can inject bad data into LLM workflows or extract sensitive procedures if guardrails are weak [185]. This makes sandboxing and strict red-teaming essential before deployment.

2) PRIVACY AND DATA PROTECTION

LLMs processing occupant data must respect privacy norms. Recent works address this through secure modeling, decentralized learning, and cryptographic storage.

- *Privacy-Preserved Thermal Modeling*: Introduces a privacy-aware parameter estimation approach that removes personal identifiers from HVAC and thermal time series, enabling aggregation without occupant-level exposure [180].
- *Blockchain for HVAC Data Security*: Proposes a decentralized blockchain framework to protect HVAC usage data in building energy systems. This ensures that LLMs using such data for control prediction cannot link it to individuals [179].
- *Federated Learning Energy Forecasting*: A federated learning framework combined with local LLMs has been proposed to predict short-term household energy consumption without compromising user privacy. The system analyzes local data (e.g., watt usage, active devices) and provides personalized recommendations via on-device LLMs, ensuring no personal data is transmitted during training. This method not only safeguards sensitive information but also promotes user trust and energy-efficient behavior [178].
- Smart-meter traces can reveal details like occupancy, appliance types, and daily routines through NILM.

Utilities have reported risks of misuse, such as profiling tenants or unfair tariff adjustments [186]. LLM assistants that analyze such data should be treated under the same privacy risk model.

- A practical way to reduce these risks is intermittently differential privacy with battery-based load hiding. Charge and discharge patterns inject calibrated noise while keeping costs low, creating a protection layer before LLM analytics [187].
- For household-level forecasting, federated learning with secure aggregation and differential privacy has been shown to work for PV production. This keeps raw data on site while maintaining accuracy and is directly compatible with edge or on-device LLMs [188].

3) BIAS, FAIRNESS, AND MISINFORMATION

LLMs, when applied to time-series or planning tasks, can generate misleading outputs due to training biases or lack of uncertainty calibration. This leads to overconfident or unfair predictions across populations and geographies.

- *Bias-Aware Forecasting for DERs*: Assesses and mitigates how LLMs may favor certain regions or grid profiles in distributed energy resource (DER) forecasting. It uses bias quantification to adjust predictions [145].
- *Hallucination-Controlled Forecasting*: Evaluates LLMs in judgmental forecasting for renewables and finds that domain-anchored prompts reduce hallucinations. Incorporating confidence intervals further prevents overconfident deployment [173].
- In grid operations, hallucinated procedures or fake references can cause unsafe switching advice or mis-prioritized outage responses. Threat models show LLMs can be pushed to generate plausible but unsafe rationales or leak internal SOPs if guardrails are weak [185]. Requiring uncertainty disclosure and tracebacks (for example, data lineage or log links) helps reduce the impact of hallucinations.
- At the policy level, LLMs can amplify misinformation and social-engineering risks, such as forged dispatch memos or phishing against operators. Security studies show these are common attack types and recommend safeguards like isolation, content provenance checks, and human verification before any policy or action is carried out [189], [190].

VII. CONCLUSION

A. SUMMARY OF FINDINGS

This paper provides an elaborate description of the applications, challenges, and future direction of LLMs in energy systems. LLMs have demonstrated potential in maintenance, demand forecasting, and analysis of energy markets based on their ability to handle structured and unstructured data. Their deployment in energy systems has enabled enhanced decision-making, automation of anomaly detection, and smart grid optimization, fault diagnosis, and documentation

automation. Besides that, LLMs' evaluation demonstrates their scalability, efficiency, and adaptability, as well as their limitations, including computational demand, bias, and interpretability.

B. BROADER IMPLICATIONS

Using LLMs in energy systems has far-reaching implications in industrial and social life. Industrially, LLMs can optimize operating efficiency, reduce downtime for power infrastructure, and improve grid reliability. Their ability to execute activities such as market analysis, policy interpretation, and compliance reduces human expert workload and increases decision accuracy. At a societal level, LLMs facilitate a sustainable energy future by optimizing renewable integration, improving energy access, and supporting climate resilience measures. Facilitating ethical AI use is necessary because LLMs must be engineered with countermeasures to bias, misinformation, and data privacy attacks.

C. CLOSING REMARKS

As power systems are developing further in the direction of greater digitization and decentralization, inter-disciplinary collaboration is required in order to actualize the full potential of LLMs. Data scientists, engineers, policy-makers, and business executives must collaborate in overcoming issues of model interpretability, real-time deployment, and integration with current infrastructure. Future breakthroughs in domain-specific fine-tuning, XAI, and hybrid AI models will keep accelerating LLMs in utility for energy applications. Through collaboration and learning, LLMs can be the basis for smart energy systems that foster efficiency, sustainability, and innovation in the energy supply sector.

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